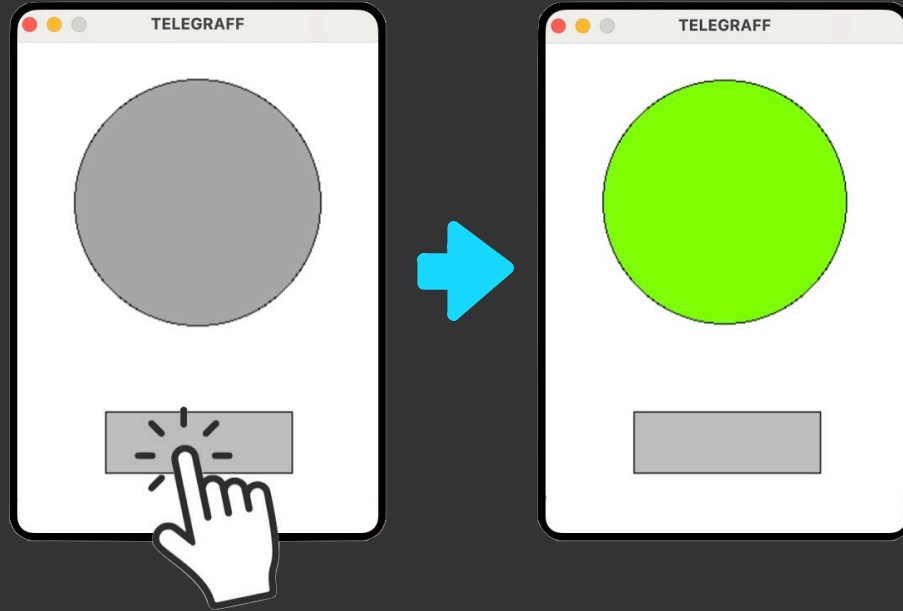


**user
events**

CSCI
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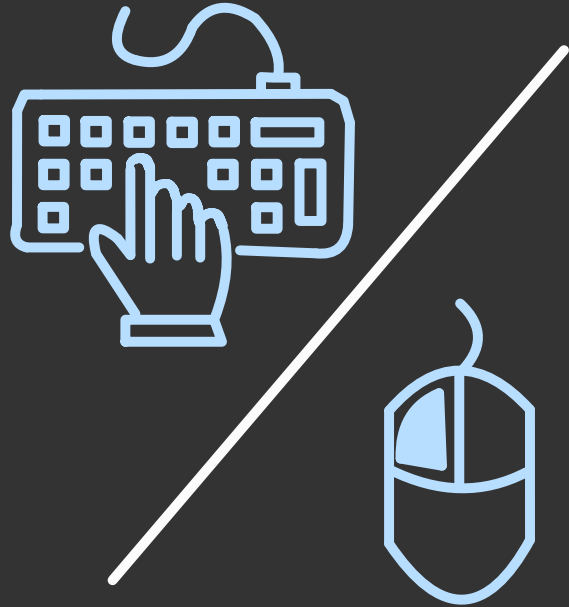
now we want to make our
morse code app **interactive**



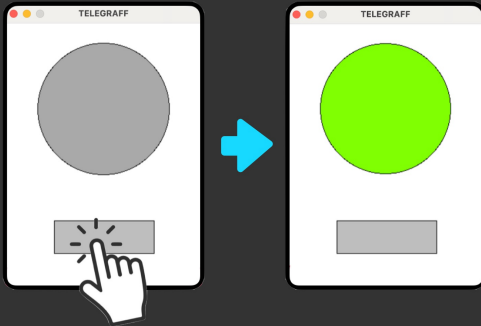
when the button
is pressed

the light
goes on

keypresses and
mouse clicks
are examples of
user events

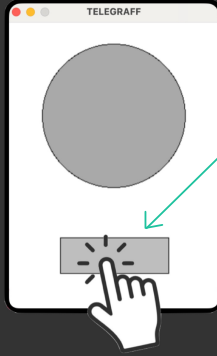


we can react to user events using the `listen_for` method of the `Window` class



```
def respond_to_mousedown(x, y):  
    lightbulb.change_fill_color("chartreuse")  
  
window.listen_for("mousedown", respond_to_mousedown)  
  
quit_now = False  
while not quit_now:  
    quit_now = window.refresh()
```

when this occurs **call this function**



when **this**
occurs

```
window.listen_for("mousedown", respond_to_mousedown)
```

this function
is called

its **arguments** are
the coordinates
where the event
occurred

```
def respond_to_mousedown(x, y):  
    if 75 <= x <= 225 and 300 <= y <= 350:  
        lightbulb.change_fill_color("chartreuse")
```

this allows us to conditionally
react to user events

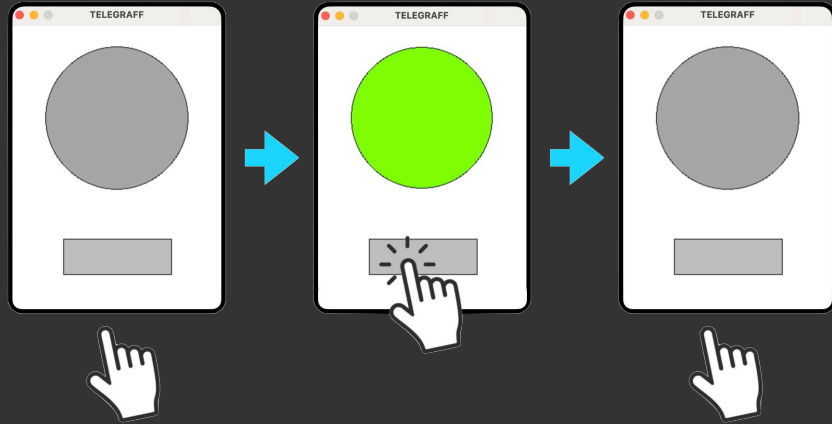
supported events

"mousedown"

"mouseup"

"keypress"

"mousemove"



```
def respond_to_mousedown(x, y):  
    if 75 <= x <= 225 and 300 <= y <= 350:  
        lightbulb.change_fill_color("chartreuse")  
  
def respond_to_mouseup(x, y):  
    lightbulb.change_fill_color("darkgray")  
  
window.listen_for("mousedown", respond_to_mousedown)  
window.listen_for("mouseup", respond_to_mousedown)
```